

MIS 443 – BUSINESS DATA MANAGEMENT

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Business Scenario and Dataset

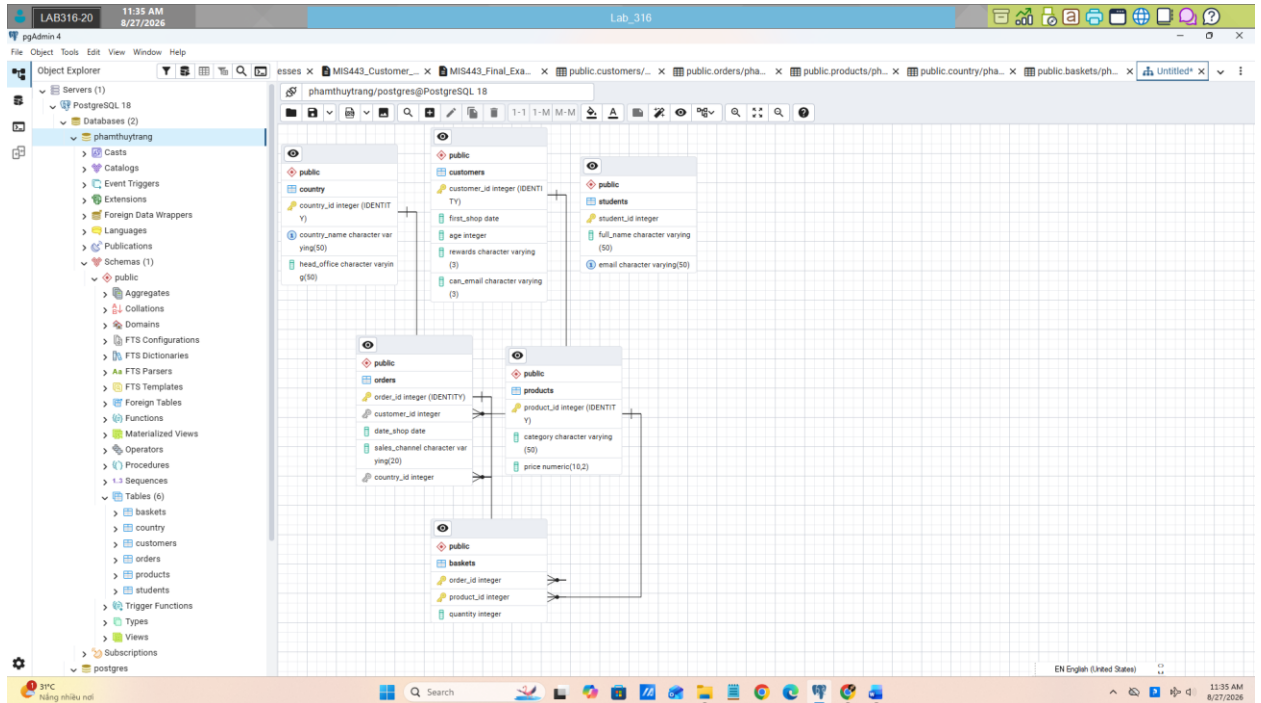
The Customer Insights database represents an international retailer selling products through retail and online channels. Management uses the database to analyse customers, orders, markets, product categories, and sales performance.

The database contains five tables:

- customers: customer age, first purchase, rewards membership, and email permission;
- orders: order date, customer, sales channel, and country;
- country: country name and head-office location;
- products: product category and unit price;
- baskets: products and quantities included in each order.

Key relationships:

- One customer can place many orders.
- One country can have many orders.
- Orders and products have a many-to-many relationship through baskets.



Question 1 – Database Setup (10 marks)

(a) Load the database – 5 marks

Create a PostgreSQL database using your full name in lowercase, without spaces or Vietnamese diacritics. Connect to the database and execute:

MIS443_Customer_Insights_PostgreSQL.sql

Confirm that country, customers, orders, products, and baskets are available in the public schema.

(b) Create the student table – 5 marks

Create public.students with:

- student_id: exactly 10 characters and primary key;
- full_name: required;
- email: required and unique.

Insert your actual information and display the inserted record.

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pgAdmin 4

Object Explorer

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Query

```
-- Question 1(b): Create public.students.
CREATE TABLE public.students(
51 student_id INTEGER PRIMARY KEY,
52 full_name VARCHAR(50) NOT NULL,
53 email VARCHAR(50) NOT NULL UNIQUE
54 );
55
56
57
58 -- Question 1(b): Insert your actual information.
59 -- Your answer here
60
61
62 -- Question 1(b): Display the inserted record.
63 SELECT * FROM public.students;
64 -- Expected result: one row containing the student's actual information.
65 /*
66 QUESTION 2 - CUSTOMER PROFILE (10 marks)
67
68 The Marketing Manager wants to understand customers who permit marketing
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 40 msec.

Total rows: Query complete 00:00:00.040

EN English (United States) LF Ln 51, Col 1

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Query

```
Insert your actual information and display the inserted record.
*/
-- Question 1(b): Create public.students.
-- Question 1(b): Insert your actual information.
-- Question 1(b): Display the inserted record.
SELECT * FROM public.students;
-- Expected result: one row containing the student's actual information.
/*
QUESTION 2 - CUSTOMER PROFILE (10 marks)
The Marketing Manager wants to understand customers who permit marketing
emails. Calculate their average age, name the result average_age, and round it
```

Data Output Messages Notifications

student_id	full_name	email
232200158	Pham Thuy Trang	trang.pham.bbs2@lelu.edu...

Showing rows: 1 to 1 Page No: 1 of 1

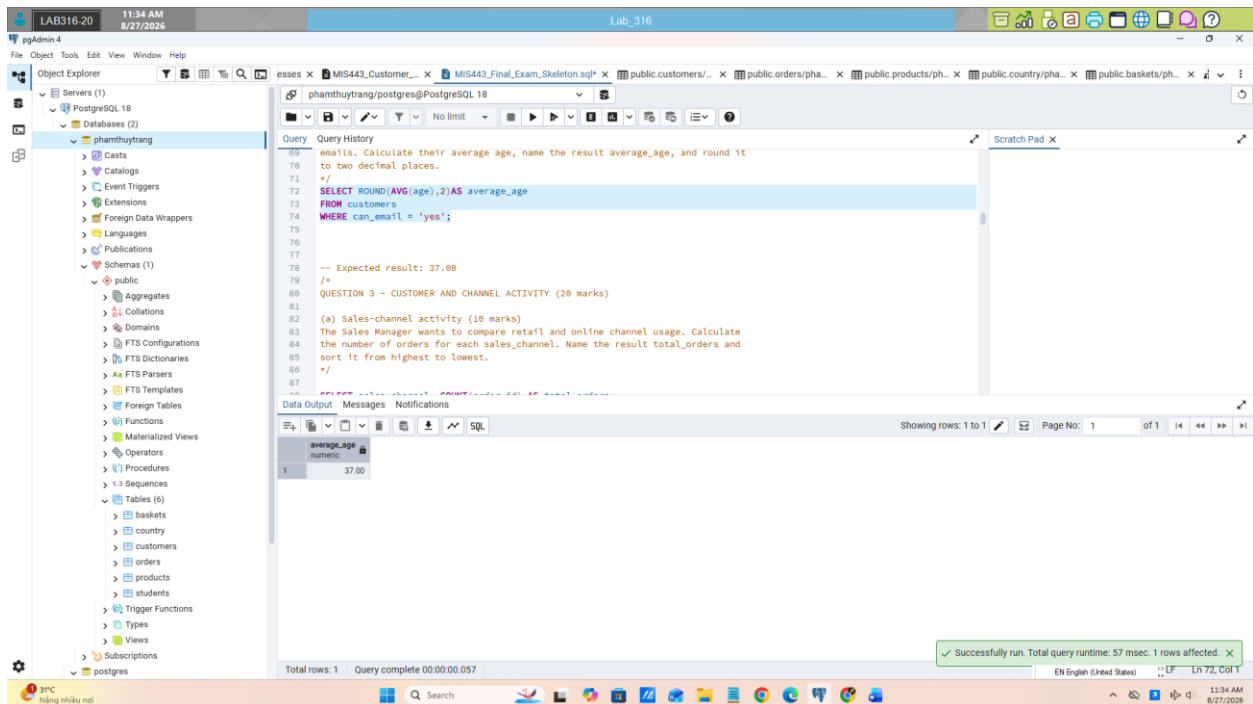
Total rows: 1 Query complete 00:00:00.055

EN English (United States) LF Ln 59, Col 1

31°C hắng nhiều nơi

Question 2 – Customer Profile (10 marks)

The Marketing Manager wants to understand customers who permit marketing emails. Calculate their average age, name the result `average_age`, and round it to two decimal places.



The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'public' schema and various tables. The main query editor window shows the following SQL query:

```
69 -- email. Calculate their average age, name the result average_age, and round it
70 -- to two decimal places.
71 */
72 SELECT ROUND(AVG(age),2)AS average_age
73 FROM customers
74 WHERE can_email = 'yes';
75
76
77 -- Expected result: 37.00
78 /*
79 QUESTION 3 – CUSTOMER AND CHANNEL ACTIVITY (20 marks)
80
81 (a) Sales-channel activity (10 marks)
82 The Sales Manager wants to compare retail and online channel usage. Calculate
83 the number of orders for each sales_channel. Name the result total_orders and
84 sort it from highest to lowest.
85 */
86
87
```

The Data Output pane shows the result of the query:

average_age
37.00

The status bar at the bottom indicates: "Successfully run. Total query runtime: 57 msec. 1 rows affected."

Question 3 – Customer and Channel Activity (20 marks)

(a) Sales-channel activity – 10 marks

The Sales Manager wants to compare retail and online channel usage. Calculate the number of orders for each `sales_channel`. Name the result `total_orders` and sort it from highest to lowest.

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pgAdmin 4

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Query

Query History

QUESTION 3 - CUSTOMER AND CHANNEL ACTIVITY (20 marks)

(a) Sales-channel activity (10 marks)

The Sales Manager wants to compare retail and online channel usage. Calculate the number of orders for each sales_channel. Name the result total_orders and sort it from highest to lowest.

```

SELECT sales_channel, COUNT(order_id) AS total_orders
FROM orders
GROUP BY sales_channel
ORDER BY total_orders DESC;

```

Data Output

sales_channel	total_orders
retail	5
online	3

Showing rows: 1 to 2 of 1

Successfully run. Total query runtime: 78 msec. 2 rows affected.

EN English (United States) LF 88, C01

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(b) Repeat customers – 10 marks

The Customer Relationship Manager wants to identify customers who placed more than one order. Display customer_id, age, and total_orders. Sort by total_orders descending and then customer_id ascending.

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pgAdmin 4

Object Explorer

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Query

Query History

(b) Repeat customers (10 marks)

The Customer Relationship Manager wants to identify customers who placed more than one order. Display customer_id, age, and total_orders. Sort by total_orders descending and then customer_id ascending.

```

SELECT
  c.customer_id,
  c.age,
  COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.age
HAVING COUNT(o.order_id) > 1
ORDER BY total_orders DESC, c.customer_id ASC;

```

Data Output

customer_id	age	total_orders
1	23	2
3	32	2

Showing rows: 1 to 2 of 1

Successfully run. Total query runtime: 65 msec. 2 rows affected.

EN English (United States) LF 107, C01

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Question 4 – Order and Country Analysis (20 marks)

(a) Order-monitoring report – 10 marks

Create an order report showing order_id, date_shop, sales_channel, customer age as customer_age, and country_name. Sort by the newest order date and then by order_id.

(b) Country performance – 10 marks

Calculate the number of orders for each country, including countries with no orders. Display country_name and total_orders. Sort by total_orders descending and then country_name alphabetically.

The screenshot shows the pgAdmin 4 interface with a SQL query editor and a data output window. The query is as follows:

```
154  
155 /*  
156 (b) Country performance (10 marks)  
157 Calculate the number of orders for each country, including countries with no  
158 orders. Display country_name and total_orders. Sort by total_orders descending  
159 and then country_name alphabetically.  
160 */  
161  
162 SELECT country_name, COUNT(o.order_id) AS total_orders  
163 FROM country c  
164 JOIN orders o ON c.country_id = o.country_id  
165 GROUP BY c.country_name  
166 ORDER BY total_orders DESC, country_name;  
167  
168  
169  
170  
171 /* Expected rows:  
172 UK | 5  
173 China | 2  
174 USA | 1  
175 */
```

The data output window shows the following results:

country_name	total_orders
UK	5
China	2
USA	1

Question 5 – Product and Revenue Analysis (20 marks)

(a) Category performance – 10 marks

For each product category, calculate total_quantity and total_revenue. Revenue equals price multiplied by quantity. Round revenue to two decimal places and sort from highest to lowest revenue.

The screenshot shows the PostgreSQL Enterprise Console interface. On the left, the Object Explorer displays the database structure, including the 'public' schema. The main query editor displays the following SQL query:

```

177
178
179 /*
180 QUESTION 5 - PRODUCT AND REVENUE ANALYSIS (20 marks)
181
182 (a) Category performance (10 marks)
183 For each product category, calculate total_quantity and total_revenue. Revenue
184 equals price multiplied by quantity. Round revenue to two decimal places and
185 sort from highest to lowest revenue.
186 */
187
188 SELECT
189     p.category,
190     SUM(b.quantity) AS total_quantity,
191     ROUND(SUM(p.price * b.quantity), 2) AS total_revenue
192 FROM baskets b
193 JOIN products p ON b.product_id = p.product_id
194 GROUP BY category
195 ORDER BY total_revenue DESC;

```

The Data Output pane shows the results of the query:

category	total_quantity	total_revenue
vitamins	7	66.93
sports	3	37.47
food	6	25.74

The status bar at the bottom indicates: "Total rows: 3 Query complete 00:00:00.064". A green notification bar at the bottom right states: "Successfully run. Total query runtime: 64 msec. 3 rows affected."

(b) High-value orders – 10 marks

Calculate the total value of each order and return only orders worth more than 20.00. Display order_id and total_order_value, rounded to two decimal places. Sort from highest to lowest value.

Question 6 – Advanced Business Analysis (20 marks)

(a) Customer purchase recency – 10 marks

For every customer who permits marketing emails, display customer_id, age, and their latest order date as latest_order_date. Include eligible customers with no orders and sort by customer_id.

(b) Customer activity ranking – 5 marks

Rank all customers by their number of orders. Customers with equal totals must receive the same rank without ranking gaps. Include customers with no orders.

Display customer_id, age, total_orders, and activity_rank. Do not use a CTE.

(c) Channel performance using a CTE – 5 marks

Use a CTE to calculate each order's total value. Then summarise performance by sales_channel.

Display:

- sales_channel;
- total_orders;
- total_revenue;
- average_order_value.

Round monetary values to two decimal places and sort by total_revenue descending